Alex M. Cooper Jr.

cooperjralex@gmail.com | (717) 693-9424 | LinkedIn | GitHub

Professional Summary

Software Engineer with hands-on experience building and deploying AI and ML systems, designing data pipelines, and delivering mission-critical software across government and industry. Proficient in applied machine learning, full-stack and cloud-native development, and systems integration. Skilled communicator with a proven record presenting technical research to senior leadership. Holds certificates in Cloud Engineering and Cyber Security. U.S. Citizen eligible for security clearance.

Experience

LUCY Mission Intern, Software/System Engineering — June 2026 – Present
National Aeronautics and Space Administration (NASA) — *Greenbelt, Maryland*

- Built an automated end-to-end data processing pipeline to ingest, analyze, and visualize Monte Carlo simulations, reducing legacy codebase by 50% while integrating 9 new analytical functionalities.
- Developed and validated a physics-based correction algorithm to mathematically reverse ion energy straggling and scattering angle delays, achieving a 70% increase in mass resolution compared to current methods.
- Constructed structured mathematical training datasets (curve-fit parameters) to train a Multi-Layer Perceptron (MLP) Neural Network for real-time FPGA hardware corrections on mission; model currently in proposal validation testing.
- Presented research to organizational leadership via PowerPoint and poster presentations for both technical and non-technical audiences.

Lucy Student Pipeline Accelerator and Competency Enabler (L'SPACE) — January 2026 – May 2026
National Aeronautics and Space Administration (NASA) — *Remote*

- Led a team of eight engineers as Lead Systems Engineer, providing logistical and technical support for mission task solution research and design.
- Designed an engineering solution for a Lunar Surface vehicle, constructed a Preliminary Design Review, and earned skill badges in System Topics, Project Management, and NX CAD.

Capstone Software Engineer — August 2025 – December 2025
Edge Optimized Image Codec — *West Chester, Pennsylvania*

- Delivered an AI-driven image codec optimized for edge devices, compressing images to a bytestream 90% smaller than JPEG using containerized, cross-platform shell scripts and Docker; developed across Agile sprints.
- Designed and implemented a UDP-based data transfer protocol for binary file transmission supporting both static and real-time streaming across clean cloud environments.

Software Engineering Intern, Data Science — June 2025 – August 2025
Booz Allen Hamilton — *Annapolis Junction, Maryland*

- Implemented an AI model for knowledge graph (KG) completion on a 100,000-node corpus of synthetic classified documents using graph neural networks and transformer-based embeddings, outputting entity and relationship inferences with upwards of 90% accuracy.
- Built a UI to streamline KG construction, modification, visualization, and completion, reducing analyst load by 75%; containerized the full product for cloud and on-device deployment.
- Presented research and results to interns, full-time employees, leadership, and executives on a national stage, placing in the top 10% of interns.

Software Engineer — January 2025 – December 2025
Campus Connect — *West Chester, Pennsylvania*

- Engineered a full-stack web application with WebSockets for live messaging, user authentication, and file sharing, supporting over 100 students in the Computer Science department.
- Deployed in a scalable cloud environment with a CI/CD pipeline via GitHub Actions and integrated an embedded RAG system enabling an off-the-shelf LLM to answer domain-specific prompts with 90% accuracy.

Education

West Chester University of Pennsylvania — West Chester, Pennsylvania
B.S. in Computer Science; Minor in Applied Statistics. Summa Cum Laude | Dean's List: Spring 2023, Spring 2024 - Spring 2026 | Honor Societies: Alpha-Alpha-Alpha, Upsilon-Pi-Epsilon | Certificates: Cloud Engineering, Cyber Security. GPA: 3.79. — *May 2026*

Skills

Languages: Python, JavaScript, C++, C, C#, Java, MATLAB, IDL, HTML, CSS, Angular

Libraries & Frameworks: scikit-learn, NumPy, Pandas, Matplotlib, networkX, FastAPI, React, Next.js, Node.js, Springboot

Tools & Platforms: AWS, Docker, Kubernetes, GitHub Actions, Terraform, Helm, CloudLab, Wireshark, SAS, SPSS, gdb, Oracle VirtualBox, Arduino IDE